

weights to zero and re-normalizing the selected weights. The MoE output is:

$$Y = \sum_{i=0}^3 w_i \cdot \text{Expert}_i(X), \quad (4)$$

followed by a 1×1 projection and a residual connection. This sparse routing mechanism operates at the feature-map level via global average pooling. It allows the network to adaptively emphasize the most relevant morphological expert branches for each input region. Consequently, both accuracy and computational efficiency are improved compared to uniform convolutions.

The BMSC2f block replaces the standard C2f bottleneck in both the backbone and neck of BMS-YOLO, integrating FDDE at the input stage and MorphSparseMoE within each bottleneck iteration. The directional convolution design draws inspiration from recent work on $1 \times 5/5 \times 1$ anisotropic kernels [23] and multi-directional residual structures [24]. Optimized C2f variants with directional convolution [23] have also shown improved performance for slender crack features.

D. LIGHTWEIGHT MULTI-SCALE FUSION: SPPF-CSPC AND ECA ATTENTION

We replace the standard SPPF module in YOLOv8 with a lightweight SPPF-CSPC variant that adopts a cross-stage partial connection (CSP) design. The SPPF-CSPC structure with cross-stage partial pooling [17] has been shown to enhance multi-scale feature aggregation for road distress detection. The input feature is split into two branches. The first branch undergoes a 1×1 convolution followed by a 3×3 convolution, and then n ($n = 3$) repeated max-pooling operations with kernel size 5. Outputs from all pooling stages are concatenated and projected by a 1×1 convolution. The second branch is a shortcut 1×1 convolution. The two branches are concatenated and passed through a final 1×1 convolution. This design enlarges the receptive field with minimal parameter increase, since pooling operations are parameter-free. It enhances multi-scale feature aggregation, which is particularly beneficial for detecting both thin, elongated cracks and large potholes within the same aerial image.

At the output of both the modified C2f blocks and the SPPF-CSPC structure, we incorporate the Efficient Channel Attention (ECA) module to refine channel-wise feature representation. Unlike the Squeeze-and-Excitation (SE) block that employs dimensionality-reduction fully connected layers, ECA directly learns channel weights via 1D convolution with an adaptively determined kernel size.

Given an input feature map $X \in \mathbb{R}^{C \times H \times W}$, global average pooling yields a channel descriptor $\mathbf{z} \in \mathbb{R}^C$. The kernel size k is computed as:

$$k = \left\lceil \frac{\log_2(C) + 1}{2} \right\rceil_{\text{odd}}, \quad (5)$$

where C denotes the number of input channels and $\lceil \cdot \rceil_{\text{odd}}$ rounds to the nearest odd integer. The scope of local cross-

channel interaction thus scales with the channel dimension. The channel weights are then $\sigma(\text{Conv1D}_k(\mathbf{z}))$, applied element-wise to the original feature map, where σ is the sigmoid activation function. ECA introduces negligible computational overhead while consistently improving the model's discriminative power across all distress categories.

E. JOINT LOSS: WIOU AND BOX MORPHOLOGY CONSISTENCY WITH TWO-STAGE TRAINING

YOLOv8 employs CIoU loss for bounding box regression. While effective for generic objects, CIoU exhibits limitations for pavement crack detection. First, the IoU component degrades rapidly for elongated boxes with minor positional deviations. Second, zero gradients are produced for non-overlapping boxes, impeding the learning of small crack fragments. Third, the aspect ratio term provides only relative supervision without constraining absolute scale.

We replace CIoU with Wise-IoU (WIoU) [33], [34], which introduces a center-distance-based sample reweighting:

$$\mathcal{L}_{\text{WIoU}} = (1 - \text{IoU}) \cdot \exp\left(\frac{\rho^2}{\sigma^2}\right), \quad (6)$$

where IoU is the intersection-over-union between the predicted and ground-truth bounding boxes, ρ is the Euclidean distance between the center points of the two boxes, and σ is the diagonal length of the minimum enclosing rectangle covering both boxes. The exponential term is detached from the gradient to serve as a sample-dependent weighting factor. This formulation assigns higher loss weights to predictions with larger center deviations, compelling the model to prioritize the correction of misaligned boxes.

Furthermore, we introduce an auxiliary morphology consistency loss that explicitly supervises the shape characteristics of elongated defects. Operating in log-space to balance the contributions of large and small boxes:

$$\mathcal{L}_{\text{morph}} = \delta\left(\log \frac{w_p}{h_p} - \log \frac{w_t}{h_t}\right) + 0.5 \cdot \delta\left(\log \sqrt{w_p h_p} - \log \sqrt{w_t h_t}\right), \quad (7)$$

where w_p and h_p denote the width and height of the predicted bounding box, and w_t and h_t denote the width and height of the ground-truth bounding box. The function $\delta(x) = x^2/2$ for $|x| < 1$ and $\delta(x) = |x| - 0.5$ otherwise (smooth L1 / Huber loss). This provides gradient stability for large deviations while preserving the ℓ_1 property near zero. The first term enforces aspect ratio consistency, directly addressing the elongated geometry of cracks. The second term regularizes the geometric mean of box dimensions to constrain absolute scale. The total box loss is:

$$\mathcal{L}_{\text{box}} = \mathcal{L}_{\text{WIoU}} + \lambda \cdot \mathcal{L}_{\text{morph}}, \quad (8)$$

where λ is a hyperparameter that balances the contribution of the morphology consistency loss. Our experiments identify a two-stage “strong-to-weak” schedule ($\lambda = 0.02$ for the pre-training phase, then $\lambda = 0.005$ for the fine-tuning phase) as the optimal strategy.

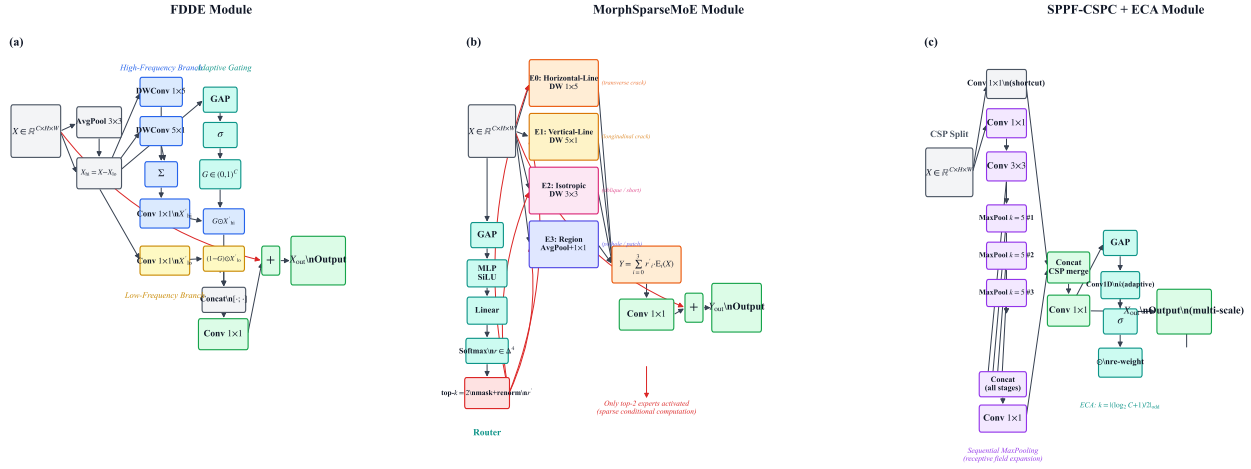


FIGURE 2. Detailed structures of the three proposed modules. (a) FDDE: the input feature is decomposed into high- and low-frequency branches; directional depth-wise convolutions (1×5 and 5×1) enhance crack-edge responses, and a gating unit $\sigma(\text{GAP}(X_{\text{high}}))$ adaptively fuses the two branches with a residual connection. (b) MorphSparseMoE: a lightweight router produces a sparse (top- $k=2$) distribution over four morphology-aware experts, whose weighted outputs are projected and added back through a residual connection. (c) SPPF-CSPC: a cross-stage partial split followed by sequential max-pooling ($k=5$) expands the receptive field at low cost, and an ECA module re-weights channels via adaptive 1D convolution.

F. INFERENCE POST-PROCESSING: TOPOLOGY-GUIDED BOX FUSION

Crack instances in UAV imagery often manifest as elongated, discontinuous structures that standard Non-Maximum Suppression (NMS) fragments into multiple detections. We propose a topology-guided box fusion strategy that treats crack fragments as topologically connected components.

We construct a graph where each detection is a node, and edges encode morphological relationships. Two crack-class boxes are considered topologically related if either of the following conditions holds: (a) their IoU exceeds a low threshold of 0.12; (b) they share the same dominant orientation (horizontal versus vertical) and their center distance d satisfies $d < 0.08 \times D_{\text{image}}$ or $d < 0.65 \times \max(s_i, s_j)$. Here, D_{image} denotes the diagonal length of the input image, and s_i and s_j denote the maximum edge lengths of the two bounding boxes being compared. Non-crack classes employ a stricter IoU threshold of 0.35. Union-Find is then applied to identify connected components. Each component is fused by a weighted combination of box coordinates (65% bounding envelope, 35% confidence-weighted average). The fused confidence is computed as the mean of original confidences, augmented by a small bonus for multi-box groups.

This fusion strategy effectively consolidates fragmented crack predictions while preserving the spatial integrity of distinct distress instances, following the principle of topology-guided Union-Find box fusion [36].

IV. EXPERIMENTS

A. EXPERIMENTAL SETUP

1) Dataset

We evaluate the proposed method on the UAV-PDD2023 dataset [37], a publicly available benchmark for pavement distress detection from unmanned aerial vehicle (UAV) im-

agery. The original dataset is hosted on Zenodo (record ID: 8429208) [37]. It comprises 2,440 high-resolution aerial images ($5,184 \times 3,888$ pixels) captured at an approximate flight altitude of 30 m, with 11,158 bounding-box annotations in Pascal VOC format. It covers six common pavement distress categories: Alligator Crack (AC), Longitudinal Crack (LC), Oblique Crack (OC), Pothole (PH), Repair (RE), and Transverse Crack (TC).

We first convert all VOC annotations to YOLO format, then partition the 2,440 original images into three subsets in a 7:2:1 ratio (random seed = 42), yielding 1,707 training, 489 validation, and 244 test images. The validation set is used for training monitoring and hyperparameter selection; the test set serves as an independent held-out evaluation set. All reported performance figures (Tables 2–6) are obtained on the test set.

The dataset exhibits a severe class imbalance. Transverse Crack dominates (48.4% of all original annotations), whereas Alligator Crack (5.4%), Pothole (1.8%), and Repair (2.5%) are extreme minority classes. To mitigate this, we apply offline augmentation-based oversampling to the training subset for the three minority classes. Using the Albumentations library, we generate augmented samples via geometric and photometric transforms. These include horizontal flip, random 90° rotation, shift-scale-rotate, random brightness/contrast, HSV shift, and Gaussian blur. We use $\text{min_visibility}=0.3$ to filter out cropped-out instances. The original training counts (Alligator Crack: 400, Pothole: 132, Repair: 212) are each raised to approximately 1,800. This adds 4,892 augmented images to the 1,707 originals, yielding a final training set of 6,599 images and 36,944 instances. The validation (489 images, 2,254 instances) and test (244 images, 1,195 instances) sets remain un-augmented. Online data augmentation (Mosaic, Random Affine, Horizontal Flip) is applied during training to all 6,599 training